

AI for Literature Reviews



Digital Integration Teaching Initiative (DITI)

Taught by Mel Williams & Sayyara Poliferno

Prof. Tom Scott

INSH6300 | Fall 2026

Discussion

- Has anyone ever used ChatGPT or another generative AI?
- What do you think it did well or poorly?
- Were there any aspects of the responses you found odd or misleading?

Workshop Agenda

- Generative AI: Bias, Confabulation/Hallucination, and Ethics
- Countering Bias
- Claude and ChatGPT for Literature Reviews
- Other AI-Powered Literature Review Tools: Elicit and Litmaps
- Conclusion
- Materials: <https://bit.ly/fa26-scott-insh6300-multi>

Generative AI

Important questions

- How do human biases impact generative AI model outputs?
- How can we counter the weaknesses of current AI models?
- How can we integrate generative AI with other tools and practices?

Vocabulary (1/2)

- **Artificial Intelligence (AI) System:** “[A]n engineered or machine-based system or functionality that is designed, for a given set of objectives, to generate outputs such as text, images, predictions, or to make recommendations or decisions influencing human action, or real or virtual environments” (Northeastern [Policy on the Use of Artificial Intelligence Systems](#)).
- **Large Language Model (LLM):** “A large language model (LLM) is a language model trained with self-supervised machine learning on a vast amount of text, designed for natural language processing tasks, especially language generation” ([Wikipedia](#)).

Vocabulary (2/2)

- Generative AI: “Generative Artificial Intelligence (AI) refers to large learning models (LLMs) that can generate high-quality text, images, and other content based on the data they were trained on, (including user-submitted text) and are designed to predict the most relevant sequence of words in response to a prompt” (Standards for the Use of Artificial Intelligence in Research)
- Bias in AI: “[A] systematic error in decision-making processes that results in unfair outcomes... [AI] bias can arise from various sources, including data collection, algorithm design, and human interpretation” (Fairness and Bias in Artificial Intelligence: A Brief Survey of Sources, Impacts, and Mitigation Strategies)

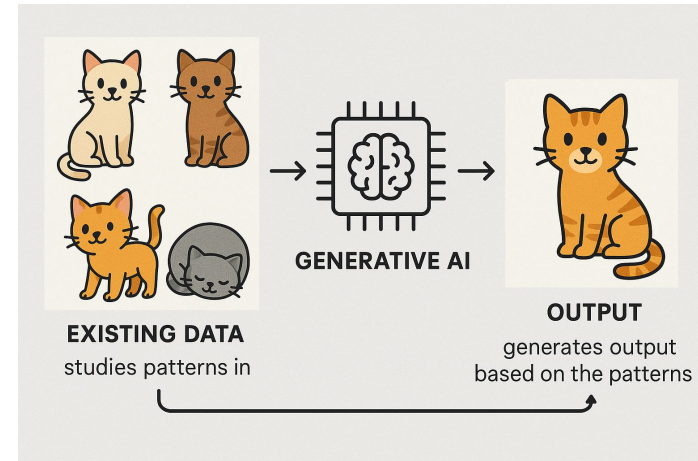
Generative AI: Introduction

AI's like ChatGPT and Claude Sonnet 5 (which generate text) and DALL-E 3 and Midjourney (which generate images) work by training on very large datasets of information and then mimicking that training data to produce “new” outputs.

Generative AI: How it Works

- Generative AIs are fundamentally based on statistics. A text-generation AI like ChatGPT generates words according to patterns learned from the training data.
- Similarly, an image-generation AI learns patterns from existing data and generates images that “looks right” but do not exist in the data. For example, AI might generate an image of a cat based on several sample images.

Generated by GPT4o
Aug 15th 2025



Example Prompt and Response

Prompt: Explain in two sentences why someone might be afraid of public speaking, in the style of Sigmund Freud. (Claude, Sonnet 5.5, Prompted on 9/29/26).

Response: “Consider the speaker upon the stage: his ego, caught between the primitive exhibitionistic wish to be admired and the harsh superego, that internalized parental authority, trembles at the prospect that the audience will judge him as the father once judged the child. The sweating palms and the faltering voice are thus nothing but the symptoms of a repressed infantile anxiety, in which the wish to be seen and the terror of being punished for that very wish collide in the unconscious..”

How the Freud Paragraph Was Generated

- A large language model analyzed Freud's existing writings and identified patterns
- These patterns are recreated in the generated text
 - Mimicking his psychoanalytic vocabulary (e.g. id, ego, superego, castration, anxiety)
 - Using Freudian logic (e.g., adult anxiety linked to childhood experiences)
 - Emulating his style of argumentation, often abstract and speculative.

Confabulation/ Hallucination in AI

Confabulation/Hallucination

- Confabulation refers to the phenomenon where large language models (LLMs) generate incorrect, nonsensical, or fabricated information, even when presented with seemingly clear and accurate prompts.
- AI confabulation is also referred to as hallucination or, more colorfully, as BS-ing.

Generative AI and “Truth”

- Text-generation AI’s aim to produce text that is grammatically correct and linguistically probable.
 - They do not understand “facts,” only patterns of word use.
- They can generate truthful text, but also frequently create confabulations/hallucinations.
 - When asked to generate citations, they can generate plausible-looking but fake sources.
 - They may link real but irrelevant sites as sources for made-up facts.
They may also invent URLs that do not work and have never worked.

News: Lawyer Cited AI Fake Cases (1/2)

- In 2026, CBS News reported that a senior lawyer in an Australian court had to apologize “for filing submissions in a murder case that included fake quotes and nonexistent case judgments generated by artificial intelligence.” (CBS, 2026)

World

**Lawyer apologizes for fake quotes,
fabricated judgments generated by AI in
murder case**

Updated on: January 27, 2026 / 6:21 AM EST / CBS/AP

 Add CBS News on Google

News: Lawyer Cited AI Fake Cases (2/2)

- The fake submissions included fabricated quotes from a speech to the state legislature and nonexistent case citations purportedly from the Supreme Court. (CBS, 2026)
- "The ability of the court to rely upon the accuracy of submissions made by counsel is fundamental to the due administration of justice," Elliott (the judge who received the hallucinated files) added.

World

**Lawyer apologizes for fake quotes,
fabricated judgments generated by AI in
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Updated on: January 27, 2026 / 6:21 AM EST / CBS/AP

 Add CBS News on Google

Hallucination: Generative AI summary of literature

- AI may inaccurately attribute concepts to individuals or omit co-contributors and earlier influences.
- It may lack citations or invent fake references.
- It may change parts of references

Hallucination Example (1/2)

Prompt: "Provide 4–5 academic citations on the writings of Shen Fu" (Claude Sonnet 5, July 27th, 2026).

1. [Helen Dunstan](#) — "If Chen Yun Had Written about Her 'Lesbianism': Rereading the Memoirs of a Bereaved Philanderer," *Asia Major* 20, no. 2 (2007): 103–122.

Correct.

2. Paul Ropp — "Between Two Worlds: Women in Shen Fu's Six Records of a Floating Life," in *Women in the History of Chinese Civilization* (1985).

This book chapter does not seem to exist.

Hallucination Example (2/2)

Prompt: "Provide 4–5 academic citations on the writings of Shen Fu" (Claude Sonnet 5, July 27th, 2026).

3. [\[Author\]](#). "A Comparative Analysis of the Reception of Four English Versions of Fu Sheng Liu Ji: Translation, Publication and International Circulation." *Translation Review* 101, no. 1 (2018): 1–17.

Missing authors
and incorrect
page numbers.

4. [Wang, Wei](#). "Pragmatic Shifts in Two Translations of *Fusheng liuji*: A Descriptive Study of Request Behaviour." *Target: International Journal of Translation Studies* 21, no. 2 (2009): 209–234.

Incorrect author.

Ethics in AI

Ethics: Generative AI “Originality”

- Some argue that all AI-generated output constitutes plagiarism and copyright infringement, since it is remixing training data that was scraped from the internet without permission from the original creators.
- Many AI companies are [facing lawsuits](#) from people whose content was used as training data without their consent.
- Some publication venues, like the *Science* journals, have made it an [official policy](#) that AI does not meet the standard for authorship and require authors to disclose use of AI.

Ethics: Generative AI Training

- AI training data is sometimes supplemented by labels (annotations) added by people (Amironesei and Díaz, [Social and Ethical Norms in Annotation Task Design](#), 2024). These labels can worsen bias in training datasets.
- People from middle and low income countries often labor in poor working conditions to annotate data for clients in high income countries. Fieldwork by Muldoon et al ([The poverty of ethical AI: impact sourcing and AI supply chains](#), 2023) revealed that workers faced traumatizing content, in addition to experiencing discrimination in the workplace and receiving low wages.

Ethics: Environment

- Training and using AI requires processing very large amounts of data, which is done in data centers
- These data centers can have a negative impact on the environment and communities
- Given their intensive energy and water demands, data centers can worsen local water scarcity and increase electricity prices
- To explore how the energy use of AI compares to other digital tasks, check out Jon Ippolito's "What Uses More" app

Ethics: AI Plagiarism Checkers

- AI plagiarism checkers, which claim to identify whether text is human or AI generated, have the potential for false positives (identifying human texts as AI).
- False positives are especially likely for texts by writers for whom English is not their first language or for writers who have autism, ADHD, dyslexia, or related neurodivergence.
- To make things worse, writers can often reduce their “AI score” by using an AI to reword their essays.

What You Can Do

- By using [smaller models](#), where possible, and running fewer prompts you may be able to reduce your contribution to AI's environmental impact.
 - To learn more about energy consumption by AI and machine learning, check out the [ML Energy Initiative](#), a collaboration among computer scientists at multiple universities, the [AI Energy Score](#), and [CodeCarbon](#).
- Public awareness can help pressure companies to adopt good practices. For more information on fair labor in AI, see the report [AI for Fair Work: From principles to practices](#) by [Fairwork](#)

Bias in AI

What is AI bias?

- Training data for AI often contains biases that are then reproduced in AI responses, potentially exacerbating harmful and unjust outcomes. Here, we refer to this as ‘AI bias’. However, there is not a single authoritative definition of AI bias.
 - For example, when AI was used to summarize medical notes, “Google’s AI tool Gemma described men’s health issues with terms like ‘disabled,’ ‘unable,’ and ‘complex’ significantly more often than women’s, who were often framed as more independent despite similar needs, an alarming gender bias trend.” (Source: crescendo.ai)
- What is the ideal “unbiased” scenario? Is an “unbiased” scenario possible?

Bias in AI Overview

- AI training data reflects the injustices and biases of the society that produced it.
- These biases can be amplified when they are input as training data into an AI, because they seem to be the “right” answers (Dwivedi et al, [Breaking the Bias: Gender Fairness in LLMs Using Prompt Engineering and In-Context Learning](#), 2023).

Bias in Source Summarization

Prompt: Write two sentences about the greatest discoveries in the social sciences and humanities and who made those discoveries. (Claude, Sonnet 5.5, 9/29/26.)

Among the greatest discoveries in the social sciences and humanities are the unconscious mind, which Sigmund Freud brought into systematic study, and the role of scarcity and exchange in human behavior, which Adam Smith and later economists such as John Maynard Keynes shaped into modern economic thought. Equally transformative were Ferdinand de Saussure's insight that language is a system of signs, which laid the groundwork for modern linguistics and structuralism, and Émile Durkheim's demonstration that social facts, such as suicide rates, can be studied empirically as products of social forces rather than individual psychology alone.

Critical examination of the AI response

What kind of theorists were included in the response response about greatest discoveries? What do you notice about this selection of people?

- Sigmund Freud
- Adam Smith
- John Maynard Keynes
- Ferdinand de Saussure
- Émile Durkheim

Bias in Source Collection (1/3)

Prompt: Provide a bibliography of 10 academic sources on climate change. Model: Claude, Sonnet 5.5, Prompted on 8/9/26. Sources:

- [Arrhenius, S.](#) (1896). On the influence of carbonic acid in the air upon the temperature of the ground. *Philosophical Magazine and Journal of Science*, 41(251), 237–276.
- [Manabe, S., & Wetherald, R. T.](#) (1967). Thermal equilibrium of the atmosphere with a given distribution of relative humidity. *Journal of the Atmospheric Sciences*, 24(3), 241–259.
- [Keeling, C. D., et al.](#) (1976). Atmospheric carbon dioxide variations at Mauna Loa Observatory, Hawaii. *Tellus*, 28(6), 538–551.

Bias in Source Collection (2/3)

Prompt: Prompt: Provide a bibliography of 10 academic sources on climate change. Model: Claude, Sonnet 5.5, Prompted on 8/9/26. Sources:

- [Hansen, J. et al.](#) (1981). Climate impact of increasing atmospheric carbon dioxide. *Science*, 213(4511), 957–966.
- [Parmesan, C., & Yohe, G.](#) (2003). A globally coherent fingerprint of climate change impacts across natural systems. *Nature*, 421(6918), 37–42.
- [Oreskes, N.](#) (2004). The scientific consensus on climate change. *Science*, 306(5702), 1686.
- [Cook, J. et al.](#) (2013). Quantifying the consensus on anthropogenic global warming in the scientific literature. *Environmental Research Letters*, 8(2), 024024

Bias in Source Collection (3/3)

Prompt: Provide a bibliography of 10 academic sources on climate change. **Model:** Claude, Sonnet 5.5, Prompted on 8/9/26. Sources:

- [Steffen, W., et al.](#) (2018). Trajectories of the Earth System in the Anthropocene. *Proceedings of the National Academy of Sciences*, 115(33), 8252–8259.
- [Edwards, P. N.](#) (2010). *A Vast Machine: Computer Models, Climate Data, and the Politics of Global Warming*. MIT Press.
- [IPCC](#) (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.

Types of Bias (1/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Geographic/ language bias	Of the 11 authors cited, 10 had only institutional affiliations in the US/ UK/Canada/Australia (or were historical scientists working in Europe).	Training data may over-represent English-language and American-centric scholarship
Longevity bias	Only 2 of the 12 sources were published within the last 10 years. A long view might be useful, but it is provided even though it is not what the prompt specifically requested.	Older sources may appear more often in training data, leading to their overrepresentation in results

Types of Bias (2/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Gender bias	Out of 11 cited authors, only 2 are women.	Historical exclusion of women from scholarship may influence disparities in training data, leading to biased results

Countering Bias

Countering Bias: Prompts and Feedback

- Bias can be partly countered through careful prompt design and feedback (Dwivedi et al, [Breaking the Bias: Gender Fairness in LLMs Using Prompt Engineering and In-Context Learning](#), 2023). However, these methods do not remove bias intrinsic in the model (Shin et al, [Can Prompt Modifiers Control Bias? A Comparative Analysis of Text-to-Image Generative Models](#), 2024)
- Practices for countering bias
 - Identify gaps or inconsistencies in generative AI responses.
 - Use additional inquiry and prompt revision to help fill in gaps.
 - Double check responses with other sources.

Women Scholars

Prompt: Write two sentences on the most influential female writers of the 18th century. (Model: Claude, Sonnet 5.5, Prompted on 9/29/26)

Among the most influential female writers of the 18th century, Mary Wollstonecraft stands out for *A Vindication of the Rights of Woman* (1792), a foundational text of feminist thought, while Frances Burney and Jane Austen's contemporaries such as Ann Radcliffe helped establish the novel as a serious literary form, with Burney's *Evelina* (1778) pioneering the novel of manners and Radcliffe's Gothic romances shaping the genre for generations. Beyond Britain, Olympe de Gouges challenged the Revolution's exclusions with her *Declaration of the Rights of Woman and of the Female Citizen* (1791), and Phillis Wheatley became the first African American woman to publish a book of poetry with *Poems on Various Subjects, Religious and Moral* (1773).

Indigenous Scholars

Prompt: Write two sentences about the most influential Indigenous writers of the 18th century. (Model: Claude, Sonnet 5.5, Prompted on 9/29/26)

Indigenous writing in the 18th century is harder to survey than the European canon, in part because much of it was produced under missionary or colonial influence and in part because oral traditions carried so much of the intellectual heritage. Among the most significant figures, Samson Occom, a Mohegan minister, published *A Sermon Preached at the Execution of Moses Paul* (1772), the first best-selling work by a Native American author, and later wrote his autobiographical *A Short Narrative of My Life* (1768), while Hendrick Aupaumut, a Mohican diplomat, produced historical writings such as his account of Mohican history and his narrative of a peace mission to the western tribes in the 1790s.

Any problems here?

Global South Writers

Prompt: Write two sentences about the most influential Global South writers of the 18th century. (Model: Claude, Sonnet 5, Prompted on 8/9/26)

Global South writing in the 18th century spans very different traditions, and much of it is documented unevenly, so any short survey is necessarily selective. Olaudah Equiano, an Igbo-born writer, published *The Interesting Narrative of the Life of Olaudah Equiano* (1789), a landmark abolitionist autobiography (though scholars debate whether he was born in Africa or South Carolina), while Ignatius Sancho, born on a slave ship and later a London grocer, left *Letters of the Late Ignatius Sancho, an African* (1782), and Ottoman poet Nedim, Bengali poet Ramprasad Sen, and Urdu poet Mir Taqi Mir each shaped the literary traditions of their regions through devotional and lyric verse. Also notable is Sor Juana Inés de la Cruz's successor generation in colonial Latin America, though she herself belongs to the late 17th century, and Tupac Amaru II's contemporaries and the Jesuit-educated writers of the Río de la Plata are worth exploring for the period.

Why do you think these writers were chosen by the model? What do they have in common?

Countering Bias: Discussion

- What groups of scholars are still excluded from these summaries?
- What other biases may they contain?
- What are biases that we cannot counter by changing prompts?
- How have you countered bias when using generative AI?

Countering Bias: Discussion

- What groups of scholars are still excluded from these summaries of the greatest discoveries in the social sciences and humanities?
- What other biases may they contain?
- What are biases that we cannot counter by changing prompts?

What You Can Do

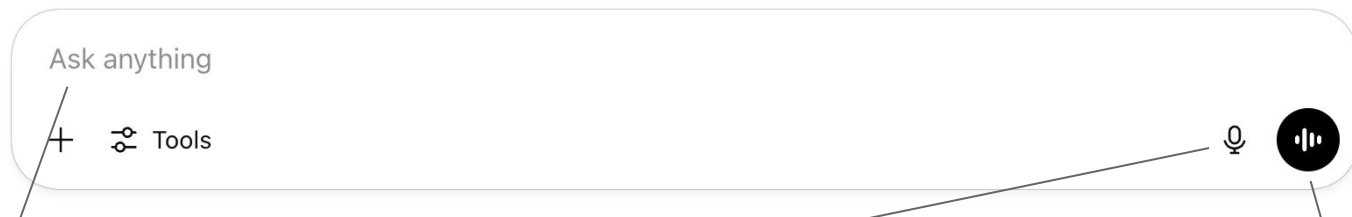
- By using [smaller models](#), where possible, and running fewer prompts you may be able to reduce your contribution to AI's environmental impact.
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- Public awareness can help pressure companies to adopt good practices. For more information on fair labor in AI, see the report [AI for Fair Work: From principles to practices](#) by [Fairwork](#)

How to use ChatGPT and Claude for literature reviews

Prompting ChatGPT (1/2)

Go to: <https://chatgpt.com/>

What's on your mind today?



A screenshot of the ChatGPT web interface. At the top, it says "What's on your mind today?". Below that is a large, rounded rectangular input box. Inside the box, on the left, is a plus sign icon and a "Tools" button. On the right side of the box are a microphone icon and a voice mode icon (a black circle with white sound waves). Three lines point from the callout boxes below to these three elements: the plus sign, the microphone icon, and the voice mode icon.

Enter prompt
in text format

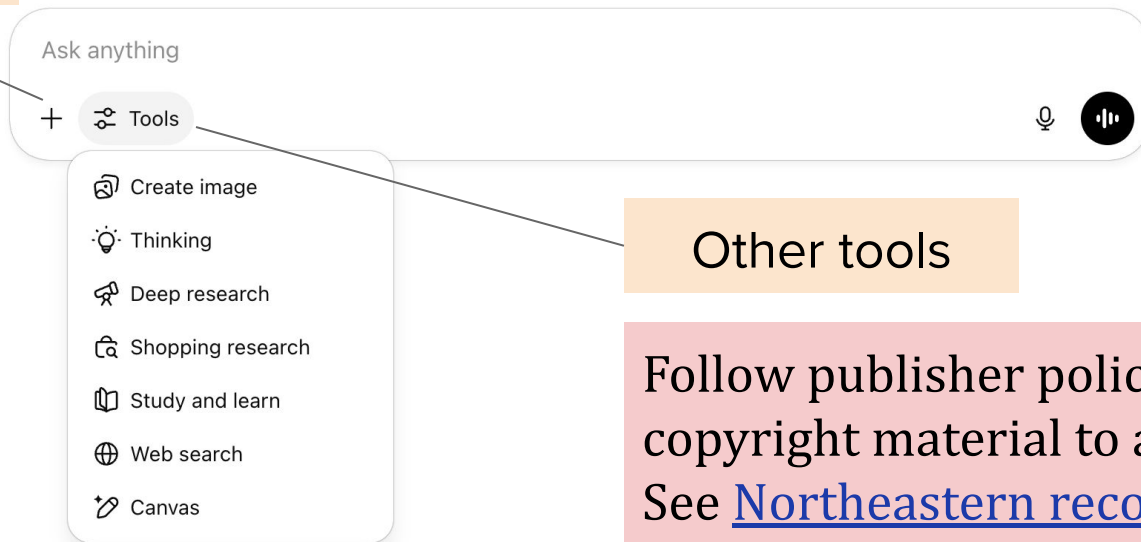
Enter prompt as speech
(transcribed automatically)

Use voice mode

Prompting ChatGPT (2/2)

Upload files

What's on your mind today?



Other tools

Follow publisher policies if uploading copyright material to an AI system. See [Northeastern recommendations](#) for more details.

ChatGPT Results

Can you provide a list of common biases in AI-generated literature reviews?

Initial prompt

Here's a clear, research-informed list of **common biases in AI-generated literature reviews**, especially those produced by large language models (LLMs). These appear across disciplines and are well-documented in recent meta-evaluations of LLM-assisted academic writing.

Response

Common Biases in AI-Generated Literature Reviews

1. Hallucinated Citations and Findings

- Fabricating articles, authors, publication years, or study results.
- Misattributing findings to real papers that never reported them.
- Citing plausible-sounding but nonexistent journals or conferences.

Enter subsequent prompts as text or speech

2. Over-Reliance on Highly Popular or High-Visibility Sources

Ask anything

+ Tools



Use voice mode

ChatGPT can make mistakes. Check important info.



Accessing Claude at Northeastern

- Northeastern has partnered with Anthropic to provide students, faculty, and staff with access to Claude
- To access Claude using your Northeastern account go to <https://claude.northeastern.edu/>
- Select **Log in with SSO**
- Login in with your Northeastern email

Prompting Claude

Enter prompt
in text format

Follow publisher policies if uploading
copyright material to an AI system.
See [Northeastern recommendations](#)
for more details.

How can I help you today?



Haiku 4.5

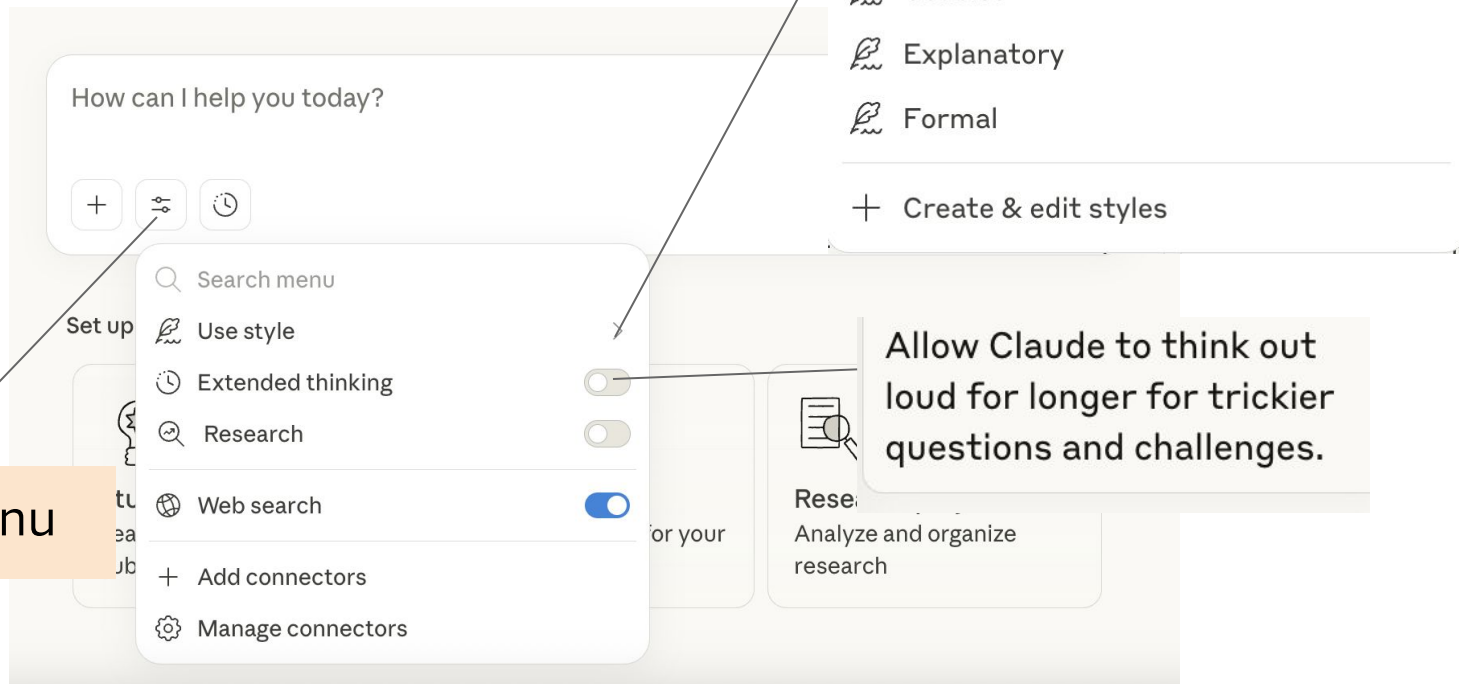


Upload files, take
screenshot, or use project

Enable extended
thinking

Select model

Prompting Claude (1/2)



Prompting Claude (2/2)

SM Can you provide a list of common biases in AI-generated literature reviews?

I'd be happy to walk through the main biases that tend to show up in AI-generated literature reviews.

Selection bias is probably the most significant. AI models are trained on text available in digital form, which means they tend to over-represent published, English-language research from well-funded institutions and underrepresent work from smaller publishers, non-English sources, and scholars from less affluent regions. This skews what gets included in the review.

Methodological bias occurs because AI tends to gravitate toward studies with clear, quantifiable results and robust methodologies—which sounds good in theory, but can actually exclude valuable qualitative research, newer methodologies, or emerging areas where rigorous standards are still being established. The AI essentially privileges a particular epistemic approach.

Recency bias appears when AI overweights recent publications simply because they're more prominent in training data or appear frequently in search results. This can make older, foundational work seem less relevant than it actually is.

↓
Social desirability bias is particularly interesting in literature reviews—the AI tends to emphasize research that reaches conclusions aligned with current academic or social

Reply...



Haiku 4.5 ▾



Initial prompt

Initial response

Enter subsequent prompts as text

What other issues do you think might arise?

Your turn

- Go to chatgpt.com or claude.ai
- Enter a literature search related prompt for a subject you are familiar with
- Identify issues in the response
 - Are any groups excluded?
 - Are the facts correct?
 - If you ask for citations, does it provide real sources?
- Revise the prompt to address the issues and generate a new response
- Evaluate the strengths and limitations of the new response

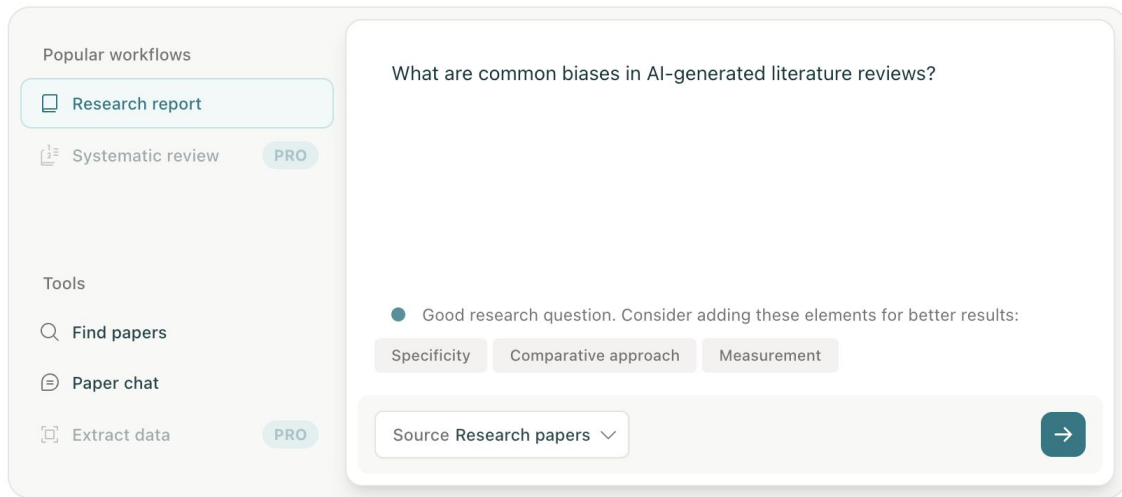
Discussion: Further Inquiry

- Was anything surprising about the responses?
- What are other ways ChatGPT and Claude could be used to counter biases in academic literature searches?
- What other types of sources could be useful for checking ChatGPT's and Claude's responses?

Other AI-Powered Literature Review Tools

Elicit

- Elicit uses AI to generate literature reports
- Provides feedback on research question



Elicit: Results (Research Report)

DECEMBER 5, 2025

What are common biases in AI-generated literature reviews?

Common biases in AI-generated literature reviews include citation bias favoring highly-cited papers, hallucination of non-existent information, reference inaccuracies, and comprehensiveness limitations that result in systematically missing relevant studies despite reasonable precision in identifying individual papers.

ABSTRACT

This review of 2 sources identified four primary types of bias in AI-generated literature reviews: citation bias, where AI systems preferentially select highly-cited studies (AI-selected papers averaged 202 citations versus 64.4 for manually-selected papers) * ; hallucination bias, involving generation of information not grounded in actual sources * ; reference inaccuracies * ; and depth/comprehensiveness bias, reflecting limited critical analysis capabilities * . These biases affect multiple stages of the systematic review process, including search and retrieval, study selection, data extraction, and synthesis.

Report

Status

✓

Gather sources

50 sources found

Details ↗

✓

Screen sources

2 sources included

Details ↗

✓

Extract data

16 data points extracted

Details ↗

✓

Generate report

Save PDF

▼

Chat

Ask anything about the results

Northeastern University
**NULab for Digital Humanities and
Computational Social Science**

56

Elicit: Results (Find Papers)

What are common biases in AI-generated literature reviews?  

 Send feedback



Sort: Most relevant

Search

Filters

+ Column

Export

UPGRADE

Save to library



Get full-texts?



☐ Paper	10 found	Summary
☐ Quantitative Measurement of Bias in AI-Generated Content: A Comprehensive Narrative Literature Review Ashish K. Saxena  International Symposium on Technology and Society, 2024, 1 citation, DOI   Abstract only		Not mentioned (the abstract discusses bias in AI-generated content but does not specify common biases in literature reviews) *
☐ Unveiling bias in artificial intelligence: Exploring causes and strategies for mitigation Yuhan Liu  Applied and Computational Engineering, 2024, 2 citations, DOI   Abstract only		Biases in AI systems include gender and race biases * ; underrepresentation of women in AI development is a confirmed bias * . Strategies for mitigation include fair datasets and increased female participation. * (Methodology: Comprehensive literature review)

Chat with papers

Edit columns



Answer



AI-generated literature reviews exhibit significant biases, particularly around gender, race, and representation, with research showing systematic limitations in bias detection and mitigation.

The evidence reveals striking patterns: [Sourojit Ghosh et al., 2025](#) found that 82% of bias research papers did not establish a clear definition of bias, and 79.9% narrowly focused on gender bias. Specifically, bias research concentrated on:

- Gender bias: 79.9% of papers
- Race/ethnicity bias: 30.2%
- Age bias: 20.6%
- Religious bias: 19.1%

[Emilio Ferrara et al., 2023](#) further emphasizes that these biases can “reproduce and amplify societal stereotypes” in generative AI content. Critically, only 10.6% of studies included recommendations for implementing bias mitigation in real-world systems, indicating a significant gap between academic research and practical application.

Litmaps

- [Litmaps](#) maps relevant literature based on citations, authors, or textual similarity
- Uses AI to find articles with similar text
- Enter search information then select seed article

Discover the world of Scientific Literature

🔍 Search by keyword, author, DOI, Pubmed ID or arXiv ID

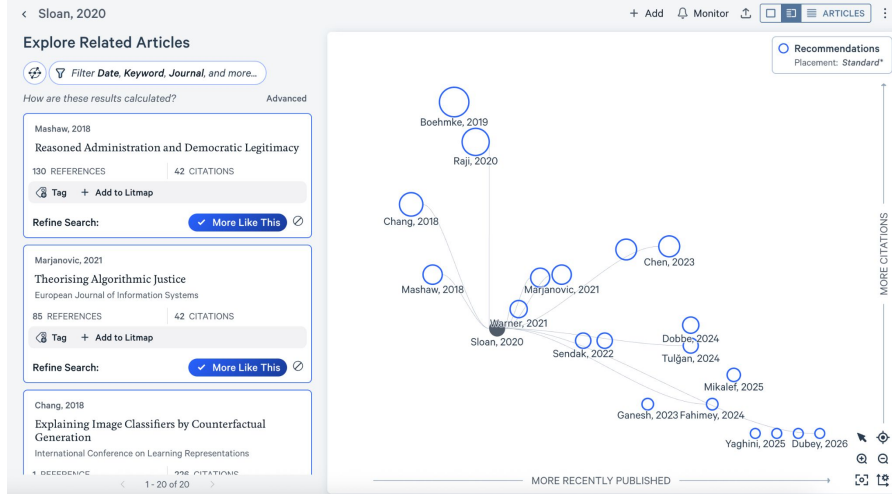
🔍 Bias, Artificial Intelligence

Eric J. Topol, 2019 High-performance medicine: the convergence of human and artificial intelligence

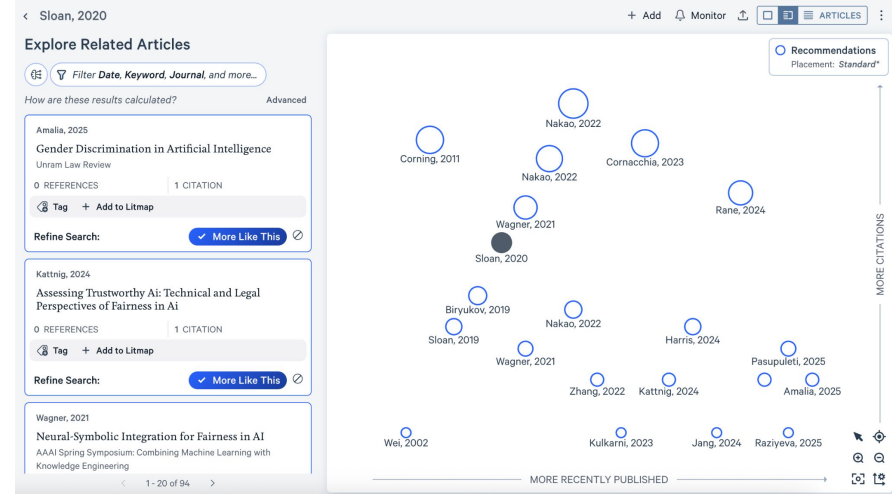
R. Sloan, 2020 Beyond Bias: Artificial Intelligence and Social Justice

Burak Koçak, 2024 Bias in artificial intelligence for medical imaging: fundamentals, detection, avoidance, mitigation, challenges, ethics, and

Litmaps: Results



Mapping by shared references and citations (does not use AI)



Mapping by textual similarity (uses AI and requires account)

Other tools

- [Scite.ai](#), finding and summarizing papers (subscribed to by Northeastern University).
- [docAnalyzer.ai](#), document analysis.
- [Consensus](#), finding papers and evaluating the extent to which their results agree.
- [Research Rabbit](#), finding and organizing papers.
- Additional [AI tools for your research](#) from Northeastern Library

Conclusion

How to use AI responsibly EVERY time

- “**Evaluate** the initial output to see if it meets the intended purpose and your needs.”
- “**Verify** facts, figures, quotes, and data using reliable sources to ensure there are no hallucinations [or confabulations] or bias.”
- “**Engage** in every conversation with the GenAI chatbot, providing critical feedback and oversight to improve the AI’s output.”
- “**Revise** the results to reflect your unique needs, style, and/or tone. AI output is a great starting point, but shouldn’t be a final product.”
- “**You** are ultimately responsible for everything you create with AI. Always be **transparent** about if and how you used AI.”
- Source from [AI for Education](#)

Main Points

- AIs, and the data used to train them, are biased
- You can design inputs to help counteract biases
- Always double check AI output
- There are multiple AI tools that may be useful at different points in your work

For Further Exploration: DITI Resources

Handouts

- [Copyright and fair use handout](#)
- [Data Ethics handout](#)
- [Data Privacy handout](#)
- [Scite handout](#)

For Further Exploration: NU Resources

Northeastern University Policy

- [Northeastern Policy on the Use of AI](#)
- [Standards for the Use of Artificial Intelligence in Research](#)
- [Generative AI in Teaching and Learning](#)

Northeastern Library

- [Research Tutorials: Generative AI](#)

Center for Advancing Teaching and Learning Through Research (CATLR)

- [AI in Teaching & Learning Across the Network](#)

Information Technology Services

- [AI at Northeastern University](#)

For Further Exploration: External (1/3)

- [Generative AI Product Tracker](#), Ithaka S+R
- [Generative AI Tools for Literature Researching](#), Oregon State University
- [Generative Artificial Intelligence \(AI\), What tools can I use for my literature review?](#), George Mason University
- [Teaching Literature Reviews in the Age of Generative Artificial Intelligence \(AI\)](#), Ozlem Tuncel, APSA Educate
- [Anthropic courses](#), Anthropic Academy

For Further Exploration: External (2/3)

- [Resisting GenAI and Big Tech in Higher Education](#), Just Sustainability Design
- [Modern Day Oracles or Bullshit Machines?](#), Bergstrom and West (2025)
- [On Being A GenAI Killjoy](#), Fernandes (2025)
- [Nothing About AI is Inevitable](#), Auslander (2025)
- [Where Cloud Meets Cement \(Data Center Report\)](#), Barakat et al (2025)
- [We are lecturers in Trinity College Dublin. We see it as our responsibility to resist AI](#), Kelly et al (2025)

For Further Exploration: External (3/3)

- [Ridicule as praxis, darling! Alex Hanna on busting “AI” hype and building the world we want](#), Shadel (2025)
- [The Cloud is Material: On the Environmental Impacts of Computing and Storage](#), Monserrate (2022)
- [Digital Revolutionary Infrastructures \(Very Encrypted\)](#), Scientist Rebellion

NULab Faculty Research (1/2)

- [“John Wihbey and Christo Wilson on Tiktok Data Espionage Concerns”](#)
- [“Alan Mislove Co-Authors Research on Discriminatory Ad Algorithms”](#)
- [“John Wihbey Weighs In On AI’s Potential to Impact the 2024 Presidential Election”](#)
- [“Tina Eliassi-Rad Co-Creates New AI Model that Predicts Human Lifespan”](#)
- [“Nabeel Gillani Interviewed by Tech Talk Podcast on AI and Education”](#)

NULab Faculty Research (2/2)

- [“John Wihbey Participates in a Panel on Content Moderation”](#)
- [“John Wihbey Comments on Google’s New ‘AI Overview’”](#)
- [“John Wihbey on the Politics of AI”](#)
- [“John Wihbey Interviewed on AI and Epistemic Risk”](#)
- [“Malik Haddad on the Regulation of AI”](#)

Thank you!

—**Developed by:** Zhen Guo, Sara Morrell, Sean Rogers, Ayah Aboelela, Johan Arango-Quiroga, Sayyara Poliferno and Mel Williams

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